

PROJECT TITLE: SMART WHEEZING PRE DETECTION AND ALERT SYSTEM

SDG: GOOD HEALTH AND WELL-BEING

Completed Field Visit? (YES)

Have a Project Idea? (YES)

STUDENT NAME	USN	DEPT	SECTION	PHONE#	ATRIA EMAIL ID
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IF YOU HAVE A TEAM , MENTION THE DETAILS BELOW:					
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FIELD VISIT DATA COLLECTION FORMAT for MULTIDISCIPLINARY PROJECT

[Phase I: Problem Identification (2nd to 3rd Semester Transition)]

1. Individual Student Information Field Visit Details

Name	USN	Branch	Contact No.	Email ID
Kishor. S	1AT24UE033	ECS	8431744417	Kishorkishors603@gmail.com

Date(s) of Visit	Sector Visited (Identify from the SDG List)	Whom did you meet? (Name, Role of the person, Phone #, Email ID (if applicable))	Address / Location where you met the person(s)	Is the person working in an organization? (Y/N)
17/9/25	Good health and well being	Mrs. Nandini	Shivaji Nagar Bengaluru	No
17/9/25	Good health and well being	Mrs. kumari	Shivaji Nagar Bengaluru	No
12/10/25	Good health and well being	Dr. Sheela Srinivas	Sheela Srinivas Clinic, Ulssor Lake, Bengaluru	Yes
12/12/25	Good health and well being	Dr. Mayuri Johari	Chinmaya Hospital, Indranagar, Bengaluru	Yes

Note: The individuals you interact with may belong to either the organised sector or the unorganised sector, for example, a fisherman, farmer, local shopkeeper, or a retired Anganwadi worker.

2. Person(s) Visited Interaction Summary (Data Collection)

Person(s) Name	Role of the Person	Sector Visited (Organized / Un-Organized)	Mode of Interaction (Personal Meetings must have photo / video evidence)
17/9/25	House wife,	Mrs. Nandini	
17/9/25	Classical dancer	Mrs. kumari	_____
Key Issues Highlighted: Miss Nandini and Miss Kumari both are suffering from the Wheezing problem.			

3. Observations & Data Collected

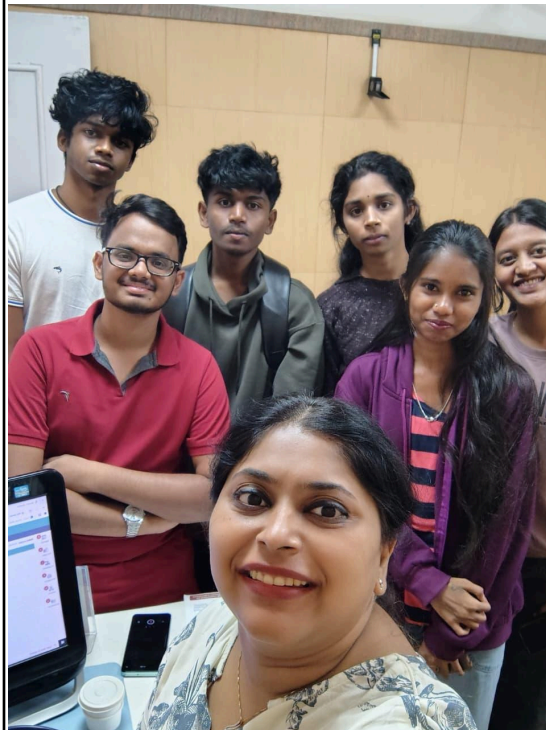
Observed Challenges/Issues	Approximate Quantitative Data	Photos/Sketches/Documents Attached
Challenge: There is no clinically standardized location for placing sensors to detect wheezing sounds. Chest placement may give better lung sounds but can be uncomfortable for continuous use, while waist placement is more comfortable but may result in weak or distorted signals. Issue Observed: Incorrect placement can lead to poor signal acquisition, noise interference, and inaccurate detection of wheezing patterns.	_____	
Challenge: As confirmed by the pulmonologist, there is currently no validated or clinically approved model that can predict wheezing episodes in advance. Issue Observed: Due to the lack of reference models or benchmark datasets, the project must rely on experimental and research-based approaches, making medical validation difficult.	_____	

Challenge:

Wheezing characteristics vary significantly between patients based on age, severity of asthma, environmental triggers, and health condition.

Issue Observed:

A single detection threshold or model cannot be applied universally, which may lead to **false alerts or missed detections**.



Note: Quantitative data for social issues refers to measurable, numerical information that helps to objectively understand the scale, impact, and urgency of a problem. (Ex: Number of people affected; Economic loss; Resource wastage; Time lost; Frequency of an issue; Health data etc.)

4. Problem Statement Draft (Tentative)

Draft Problem Statement(s)

Many people, especially asthma and wheezing patients, often experience sudden breathing problems without any early signs.

Current medical devices only detect wheezing after it starts, which is often too late to prevent an attack.

Due to this delay, patients suffer severe breathing discomfort, panic, and even emergency hospitalizations.

There is an urgent need for a system that can monitor breathing patterns continuously and predict upcoming wheezing attacks before they happen.

Note: Write a clear and short problem statement about a real issue, using what you observed during the visit and what you learned from the people you spoke to.

5. Alignment with UN Sustainable Development Goals (SDGs)

Relevant SDG(s)	Justification (How the problem aligns)
Good health and well-being	This project is important because it helps people who suffer from chronic wheezing by continuously monitoring their breathing and sending alerts to nearby clinics or hospitals. It provides immediate support during wheezing episodes, ensuring timely medical help and improving the patient's safety and quality of life

JUSTIFICATION

This project can make a strong positive social impact:

- It can save lives by alerting patients before a severe breathing attack occurs.
- It helps elderly and rural patients who may not have immediate access to doctors.
- It can reduce hospital visits and healthcare costs by preventing emergencies.
- The system is affordable and user-friendly, making it suitable for large-scale use.
- In the long term, this project can contribute to creating healthier communities and smarter healthcare networks

Product Function: The new product is a patch test intended to detect and predict attacks, possibly 30 minutes to an hour before they start. It uses sensors and a microcontroller to connect to the web (via a smartphone) for data transmission.

Need for Medical Endorsement: The primary advice is to seek guidance from Chest Physicians (Pulmonologists), as they are the best experts for endorsing and providing feedback on the product. She emphasizes that the device is a "new thing" and needs expert validation.

Reasoning: Pulmonologists specialize in cases like allergies, asthma, bronchitis, and COPD (Chronic Obstructive Pulmonary Disease).

Consultation Suggestion: The pulmonologist suggested that wheezing signals are highly dependent on proper sensor placement, with chest placement being more clinically relevant than waist placement. She emphasized that wheezing detection and prediction is a research-level problem, as no validated model currently exists.

Cost and Accessibility Concerns:

The woman highlights that the product must consider the cost factor.

She notes that individuals who need the device most, such as construction workers or those exposed to dust, are often from lower-income brackets.

She questions if this demographic can afford the product, especially since they already struggle to afford basic medication.

The student mentions the estimated budget is around ₹4,000–₹5,000.

This patch test and AI-driven system represents a significant leap forward in respiratory care by being one of the first products to offer:

Proactive Prediction: Unlike standard devices (like peak flow meters) that only measure lung capacity after symptoms appear, our technology is designed to predict an attack 30 minutes to one hour before it strikes.

Continuous, Non-Invasive Monitoring: It offers continuous monitoring via a wearable, non-invasive patch, providing real-time data that traditional periodic tests cannot capture.

Affordable, Accessible Alert System (Target Goal): The goal is to deliver crucial, potentially life-saving alerts directly to the user's smartphone (or ordinary phone), making predictive healthcare accessible outside of clinical settings.

Bridging the Gap: We aim to bridge the gap between simple detection and proactive warning, giving patients the critical time needed to use their inhalers or seek medical aid before the attack reaches its peak severity.



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